

EEE 359

Engineering Project Management

Lecture 02

Project Context

Project Environment

Projects are influenced by a **multitude of factors** which can be **external or internal** to the organization responsible for its management and execution. The important thing for the project manager is to recognize what these factors are and how they impact the project during various phases from inception to final handover, or even disposal. These external or internal influences are known as the **project context or project environment**.

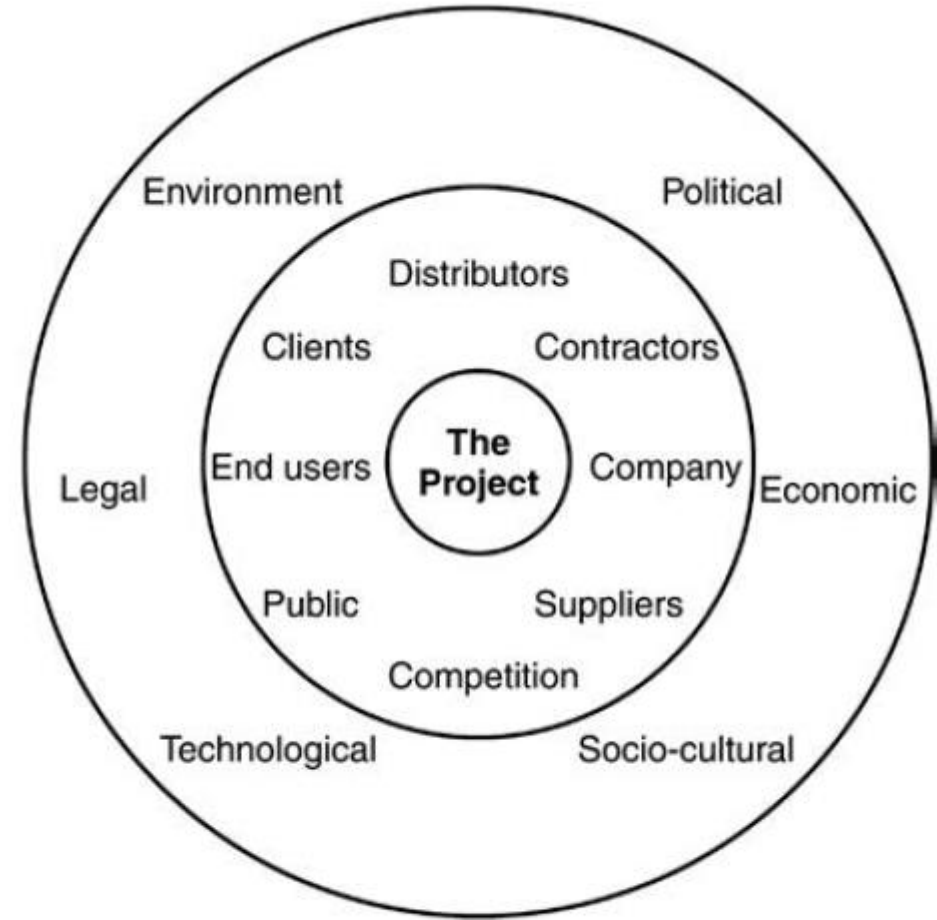
The external factors making up this environment are the client or customer, various external consultants, contractors, suppliers, competitors, politicians, national and local government agencies, public utilities, pressure groups, the end users and even the general public.

Internal influences include the organization's management, the project team, internal departments (technical and financial) and possibly the shareholders.

Project Environment

All these influences are encapsulated by the acronym PESTLE, which stands for:

- Political
- Economic
- Social
- Technical
- Legal
- Environmental



The project environment.

Important Knowledge

Phase Gate

A phase gate is held at the end of a phase. The project's performance and progress are compared to project and business documents including but not limited to:

- Project business case
- Project charter
- Project management plan
- Benefits management plan

A decision (e.g., go/no-go decision) is made as a result of this comparison to:

- Continue to the next phase
- Continue to the next phase with modification
- End the project
- Remain in the phase
- Repeat the phase or elements of it

Project Management Process Group

Five Project Management Process Groups:

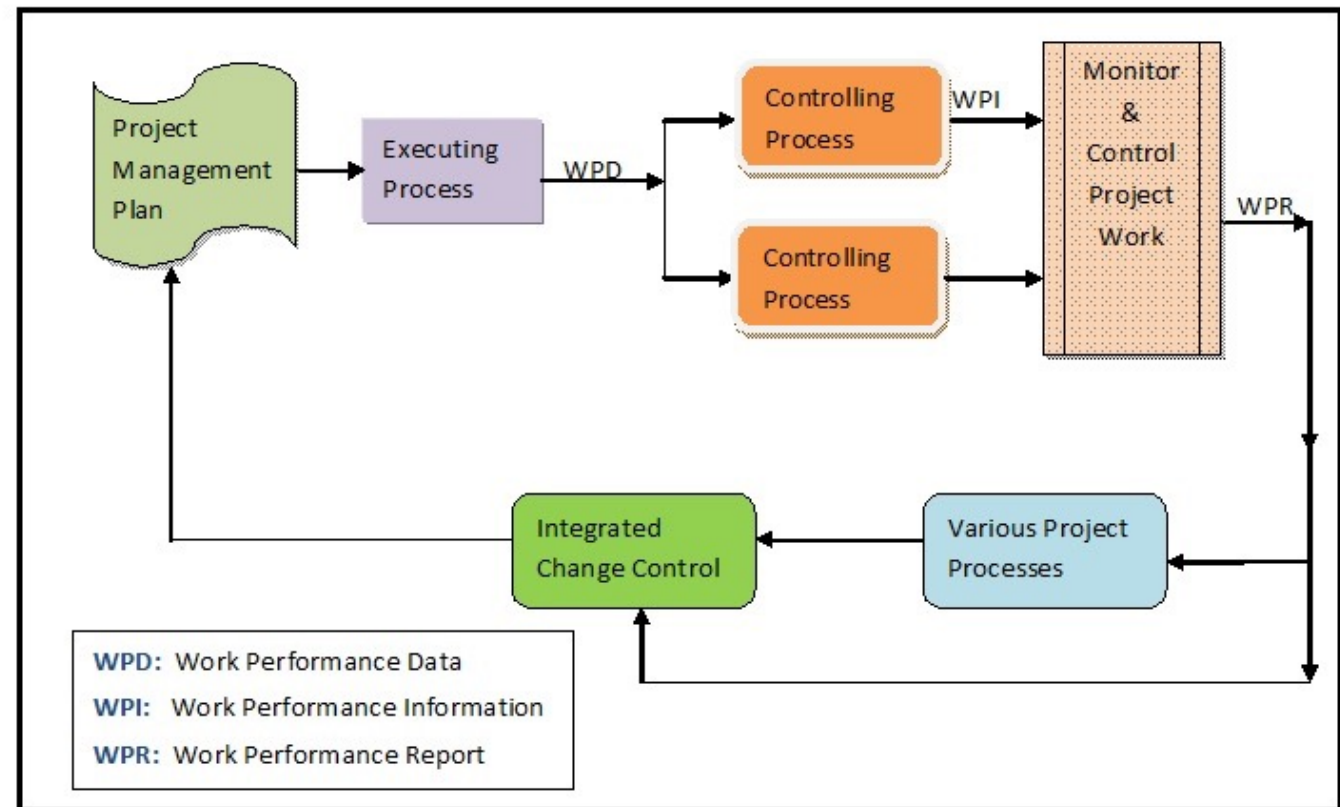
- **Initiating Process Group:** Those processes performed to define a new project or a new phase of an existing project by obtaining authorization to start the project or phase.
- **Planning Process Group:** Those processes required to establish the scope of the project, refine the objectives, and define the course of action required to attain the objectives that the project was undertaken to achieve.
- **Executing Process Group:** Those processes performed to complete the work defined in the project management plan to satisfy the project requirements.
- **Monitoring and Controlling Process Group:** Those processes required to track, review, and regulate the progress and performance of the project; identify any areas in which changes to the plan are required; and initiate the corresponding changes.
- **Closing Process Group:** Those processes performed to formally complete or close the project, phase, or contract.

Project Management Data and Information

- **Work performance data:** The raw observations and measurements identified during activities performed to carry out the project work. Examples include reported percent of work physically completed, quality and technical performance measures, start and finish dates of schedule activities, number of change requests, number of defects, actual costs, actual durations, etc. Project data are usually recorded in a Project Management Information System (PMIS) and in project documents.
- **Work performance information:** The performance data collected from various controlling processes, analyzed in context and integrated based on relationships across areas. Examples of performance information are status of deliverables, implementation status for change requests, and forecast estimates to complete.
- **Work performance reports:** The physical or electronic representation of work performance information compiled in project documents, which is intended to generate decisions or raise issues, actions, or awareness. Examples include status reports, memos, justifications, information notes, electronic dashboards, recommendations, and updates.

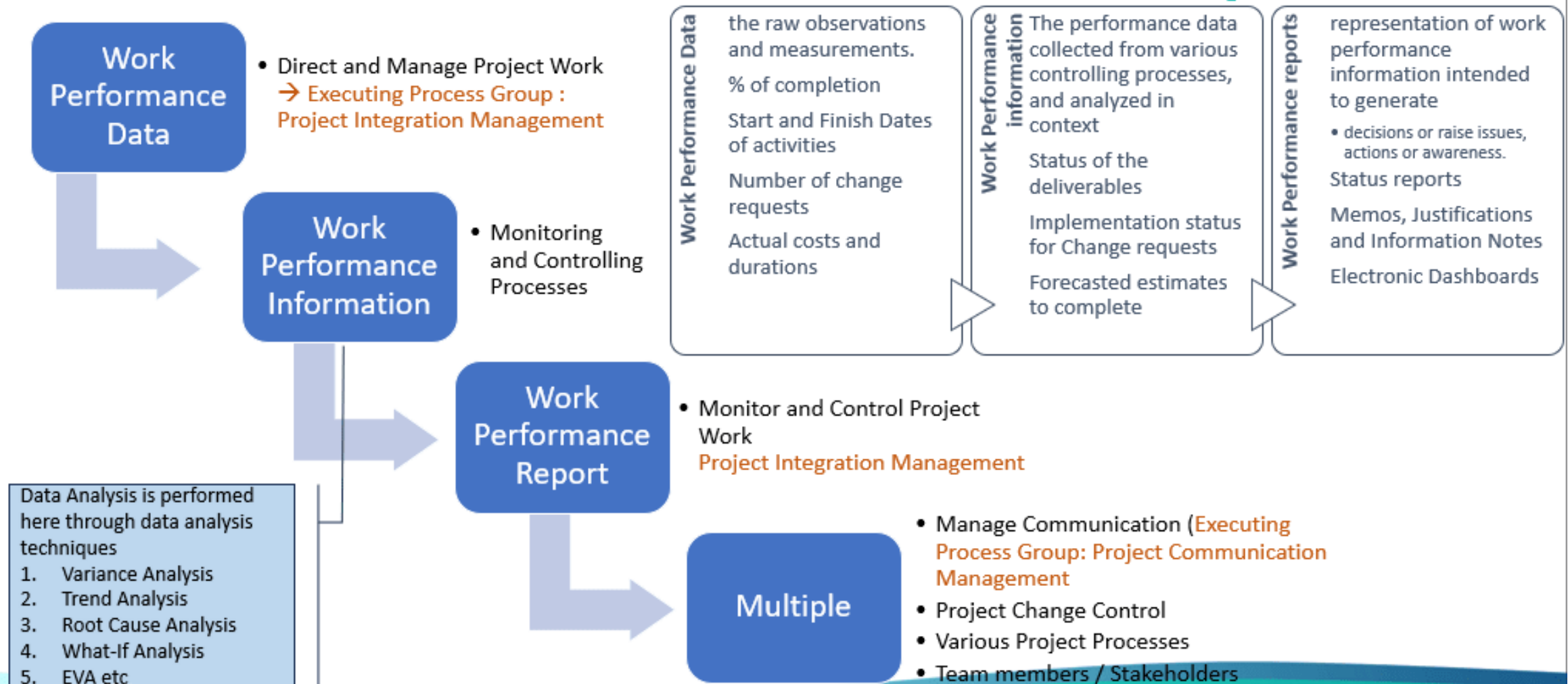
Project Management Data and Information

- The **Work Performance Data** are the initial measurements and details about activities gathered during the project work.
- This Work Performance data is analyzed to make sure it confirms to the project management plan and is assessed to what it means for the project as a whole... this result is **Work Performance Information**.
- This information is then organized into **Work Performance Reports** which are distributed to various stakeholders who need to receive and possibly act and make decisions based on the information. The reports are outputs in monitoring and control project work."



Project Management Data and Information

Work Performance Data, Information and Reports



Organizational Structure

Factors to Select

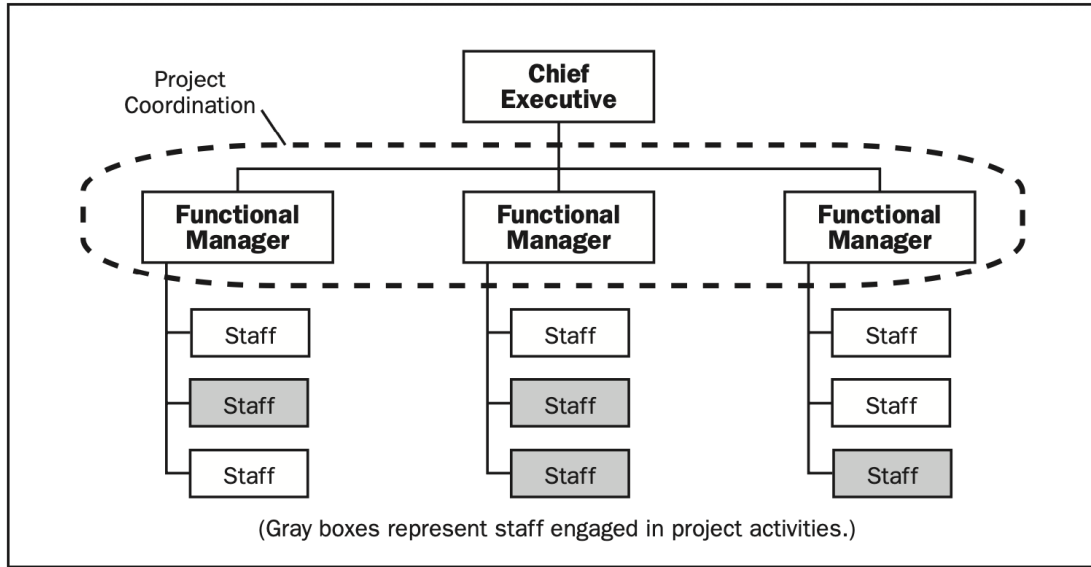
Factors to consider in selecting an organizational structure include but are not limited to:

- Degree of alignment with organizational objectives
- Specialization capabilities
- Span of control, efficiency, and effectiveness
- Clear path for escalation of decisions
- Clear line and scope of authority
- Delegation capabilities
- Accountability assignment
- Responsibility assignment
- Adaptability of design
- Simplicity of design
- Efficiency of performance
- Cost considerations
- Physical locations (e.g., colocated, regional, and virtual)
- Clear communication (e.g., policies, status of work, and organization's vision)

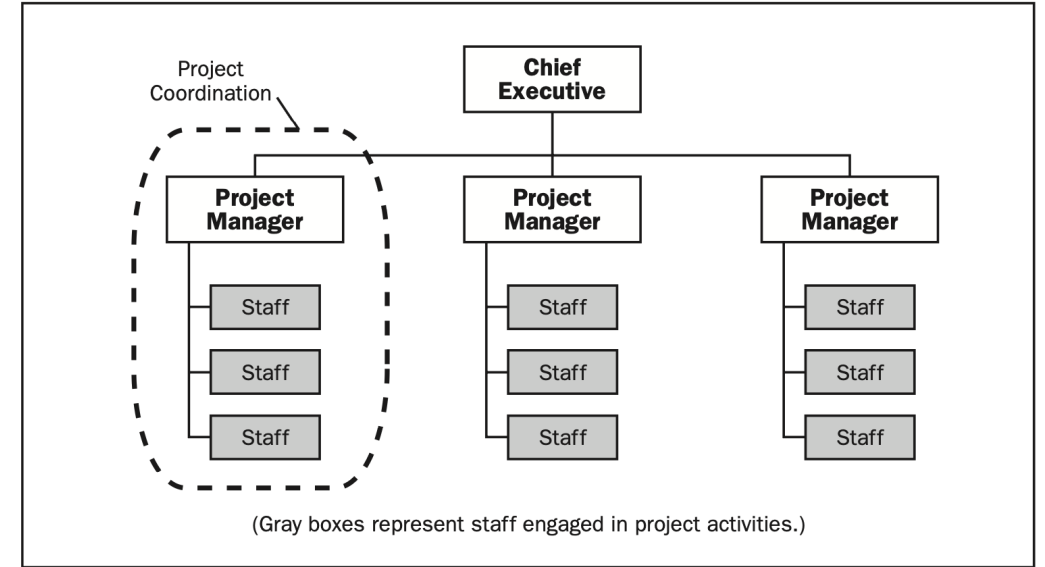
Organizational Structures

- **Organic or Simple Organization** - the simplest, includes freelancers, startups. Owner bears most responsibility.
- **Functional or Centralized Organization** - most commonly used, consists of various departments where people with similar skills are put together, structured into areas and departments, e.g. sales, marketing, finance.
- **Multi-divisional Organization** - many functional divisions with small centralization. People with different skills, are held together according to a similar product, service or geographic area. E.g. different branches of a company.
- **Matrix Organization – strong, weak or balanced** - hybrid between functional and project-oriented.
- **Project Oriented (composite or hybrid) Organization** - when an org. considers every job as a project.
- **Virtual Organization** - central organization connects to external companies (such as vendors, customers, associates) with a network connection that allows to achieve business growth and profitability. Most processes are outsourced.
- **Hybrid** - combination of the above-mentioned structure types. Responsibility, authority, and other factors vary.
- **PMO (Project Management Office)** - mixed type of organizational structure, but the project manager has the highest authority, controls the budget, and has a team completely at his disposal.

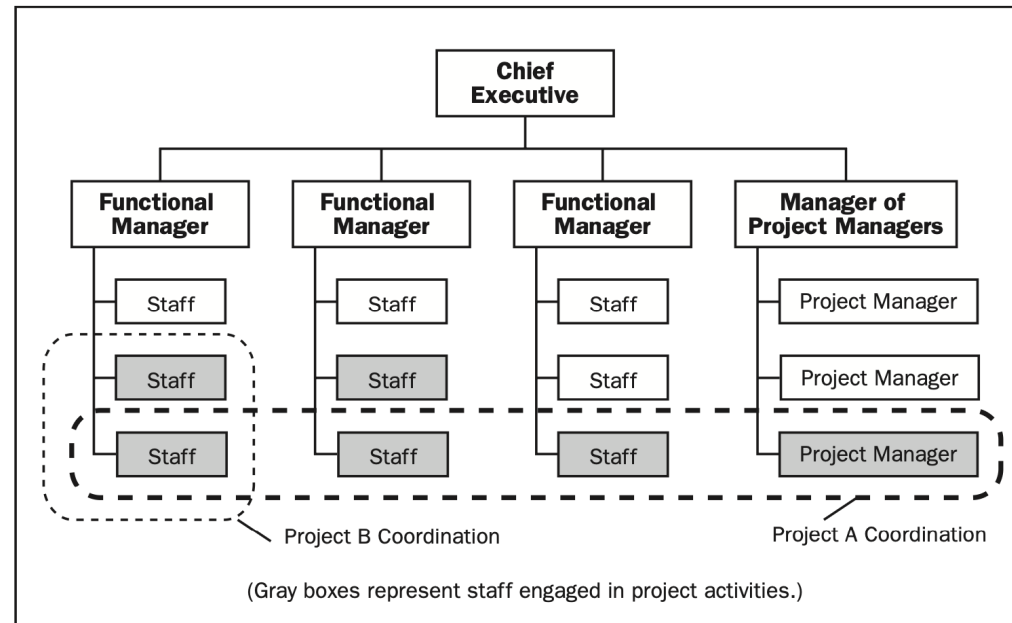
Organizational Structures



Functional Organization



Projectized Organization



Composite Organization

Organizational Structures Comparison

Organization Structure Project Characteristics	Functional	Matrix			Projectized
		Weak Matrix	Balanced Matrix	Strong Matrix	
Project Manager's Authority	Little or None	Limited	Low to Moderate	Moderate to High	High to Almost Total
Resource Availability	Little or None	Limited	Low to Moderate	Moderate to High	High to Almost Total
Who controls the project budget	Functional Manager	Functional Manager	Mixed	Project Manager	Project Manager
Project Manager's Role	Part-time	Part-time	Full-time	Full-time	Full-time
Project Management Administrative Staff	Part-time	Part-time	Part-time	Full-time	Full-time

Project Management Office

Project Management Office (PMO)

A project management office (PMO) is an **organizational structure** that standardizes the project-related governance processes and facilitates the sharing of resources, methodologies, tools, and techniques. The responsibilities of a PMO can range from providing project management support functions to the direct management of one or more projects.

The project management office may have organization-wide responsibility. It may play a role in supporting strategic alignment and delivering organizational value. The PMO integrates data and information from organizational strategic projects and evaluates how higher-level strategic objectives are being fulfilled. The PMO is the natural liaison between the organization's portfolios, programs, projects, and the organizational measurement systems (e.g., balanced scorecard).

Project Management Office (PMO)

Primary function of a PMO is to **support project managers** in a variety of ways, which may include but are not limited to:

- Managing shared resources across all projects administered by the PMO
- Identifying and developing project management methodology, best practices, and standards
- Coaching, mentoring, training, and oversight
- Monitoring compliance with project management standards, policies, procedures, and templates by means of project audits
- Developing and managing project policies, procedures, templates, and other shared documentation (organizational process assets)
- Coordinating communication across projects

Project Management Office (PMO)

Several types of PMOs in organizations. Each type varies in the degree of control and influence it has on projects within the organization, such as:

- **Supportive:** Supportive PMOs provide a consultative role to projects by supplying templates, best practices, training, access to information, and lessons learned from other projects. This type of PMO serves as a project repository. The degree of control provided by the PMO is low.
- **Controlling:** Controlling PMOs provide support and require compliance through various means. The degree of control provided by the PMO is moderate. Compliance may involve:
 - ✓ Adoption of project management frameworks or methodologies
 - ✓ Use of specific templates, forms, and tools
 - ✓ Conformance to governance frameworks
- **Directive:** Directive PMOs take control of the projects by directly managing the projects. Project managers are assigned by and report to the PMO. The degree of control provided by the PMO is high.

Role of Project Manager & Project Sponsor

Project Manager

The project manager is the person assigned by the performing organization to lead the team that is responsible for achieving the project objectives.

The PMI (Project Management Institute) talent triangle focuses on three key skill sets:

- **Technical project management:** The knowledge, skills, and behaviors related to specific domains of project, program, and portfolio management. The technical aspects of performing one's role
- **Leadership:** The knowledge, skills, and behaviors needed to guide, motivate, and direct a team, to help an organization achieve its business goals
- **Strategic and business management:** The knowledge of and expertise in the industry and organization that enhanced performance and better delivers business outcomes

Project Sponsor

The person, who is often a director of the client's organization with a special brief to oversee the project, is the **project sponsor** (sometimes also known as the project champion).

The role of the project sponsor is:

- Monitor the performance of the project manager
- Constantly ensure that the project's objectives and main criteria are met
- Ensure that the project is run effectively as well as efficiently
- Assess the need and viability of variations and agree to their implementation
- Assist in smoothing out difficulties with other stakeholders
- Support the project by ensuring sufficient resources (especially financial) are available
- Act as business leader and top-level advocate to the company board
- Ensure that the perceived benefits of the project are realized

Depending on the value, size and complexity of the project, the sponsor is a key player who, as a leader and mentor, can greatly assist the project manager to meet all the project's objectives and KPIs.

Leadership

Qualities of Leader

Research shows that the qualities and skills of a leader include but are not limited to:

- Being a visionary (e.g., help to describe the products, goals, and objectives of the project; able to dream and translate those dreams for others)
- Being optimistic and positive
- Being collaborative
- Managing relationships and conflict by
 - * Building trust
 - * Satisfying concerns
 - * Seeking consensus
 - * Balancing competing and opposing goals
 - * Applying persuasion, negotiation, compromise, and conflict resolution skills
 - * Developing and nurturing personal and professional networks
 - * Taking a long-term view that relationships are just as important as the project
 - * Continuously developing and applying political acumen

Qualities of Leader

- Communicating by:
 - * Spending sufficient time communicating (research shows that top project managers spend about 90% of their time on a project in communicating)
 - * Managing expectations
 - * Accepting feedback graciously
 - * Giving feedback constructively
 - * Asking and listening
- Being respectful (helping others retain their autonomy), courteous, friendly, kind, honest, trustworthy, loyal, and ethical
- Exhibiting integrity and being culturally sensitive, courageous, a problem solver, and decisive
- Giving credit to others where due
- Being a life-long learner who is results- and action-oriented

Qualities of Leader

- Focusing on the important things, including:
 - * Continuously prioritizing work by reviewing and adjusting as necessary
 - * Finding and using a prioritization method that works for them and the project
 - * Differentiating high-level strategic priorities, especially those related to critical success factors for the project
 - * Maintaining vigilance on primary project constraints
 - * Remaining flexible on tactical priorities
 - * Being able to sift through massive amounts of information to obtain the most important information
- Having a holistic and systemic view of the project, taking into account internal and external factors equally
- Being able to apply critical thinking (e.g., application of analytical methods to reach decisions) and identify him or herself as a change agent
- Being able to build effective teams, be service-oriented, and have fun and share humor effectively with team members

Comparison between Leadership and Management

Management	Leadership
Direct using positional power	Guide, influence, and collaborate using relational power
Maintain	Develop
Administrate	Innovate
Rely on control	Inspire trust
Focus on near-term goals	Focus on long-range vision
Ask how and when	Ask what and why
Focus on bottom line	Focus on the horizon
Accept status quo	Challenge status quo
Do things right	Do the right things
Focus on operational issues and problem solving	Focus on vision, alignment, motivation, and inspiration

Leadership Styles

- Laissez-faire (e.g., allowing the team to make their own decisions and establish their own goals, also referred to as taking a hands-off style)
- Transactional (e.g., focus on goals, feedback, and accomplishment to determine rewards; management by exception)
- Servant leader (e.g., demonstrates commitment to serve and put other people first; focuses on other people's growth, learning, development, autonomy, and well-being; concentrates on relationships, community and collaboration; leadership is secondary and emerges after service)
- Transformational (e.g., empowering followers through idealized attributes and behaviors, inspirational motivation, encouragement for innovation and creativity, and individual consideration)
- Charismatic (e.g., able to inspire; is high-energy, enthusiastic, self-confident; holds strong convictions)
- Interactional (e.g., a combination of transactional, transformational, and charismatic)

Thanks